

## Section 4: Architectural Patterns & Styles

### Introduction

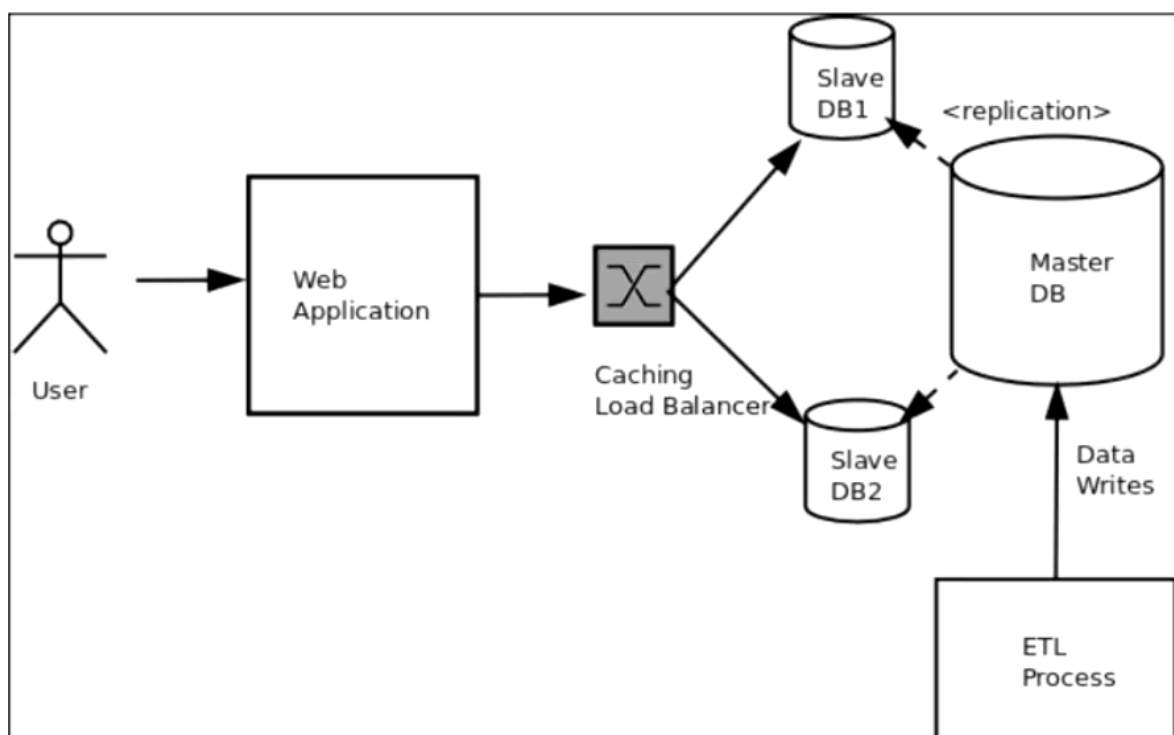
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In this section we will explore the core concept behind software architecture, focusing on the most common patterns and styles used in modern system design. By the end of this section, you will understand the different architectural approaches, their advantages and trade-offs and how to choose the right one for your system.

### What is Software Architecture?

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At its core, software architecture refers to the structure of a system, how its components are organized and how they interact with each other. This structure includes everything from data flows to module dependencies, and even the ways different parts of the systems communicate.



Architecture has a profound impact on scalability, performance and maintainability of our application. These are the pillars that ensure your system can handle growth, respond efficiently and remain adaptable over time. When designing a system it is critical to consider the following key design considerations:

- **Scalability:** It is the system's ability to handle an increasing amount of data or user traffic without compromising performance. A scalable system can grow as your needs grow.
- **Maintainability:** It is how easily the system can be updated or modified. A well-architected system is one that allows for easy updates, improvements, and bug fixes without causing a major disruption.
- **Performance:** It refers to how efficiently the system performs under load, how quickly it processes requests and responds to users, and how well it performs as the traffic increases.

Remember, the architectural choices you make directly influences the system behaviour. So every decision whether you use a monolithic or microservices approach to how you structure your databases, it will impact on how your system behaves in production.

## Section Agenda

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- Lecture 1: Introduction to Architectural Patterns
- Lecture 2: Software Architecture & Styles
- Lecture 3: Multi-tier Architecture
- Lecture 4: Microservices Architecture
- Lecture 5: Event-Driven Architecture
- Lecture 6: Summary & Practical Applications

By the end of this section, you will have a solid understanding of how key architectural patterns are used in modern system design, how to choose between them, and how to implement them effectively.